Hüseyin Felek

R&D Engineer | Industrial Automation & Process Instrumentation

Istanbul, Turkey | (+90) 505 974 84 01 | hseyin.felek@gmail.com | linkedin.com/in/hfelek

SUMMARY

Engineer with nearly 5 years in industrial automation and process instrumentation, developing products and carrying projects from requirement definition through design, development, testing and handover to production and the field. Owned a new instrumentation product end to end, developed an automation controller platform deployed on 1000+ field devices, and worked closely with customers, production, quality and field service teams and external suppliers. Hands-on with industrial communication protocols (PROFINET, Modbus, IO-Link, RS-485, 4-20 mA), PLC programming in CODESYS (IEC 61131-3), embedded software and HMI development, and production testing and quality assurance.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Delivered core network features (VLAN, IEEE 802.1X/RADIUS, ACL) for an OpenWrt-based switching and routing platform to production readiness, troubleshooting reliability issues at packet level.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Product Development and Instrumentation

- Defined product requirements and technical parameters for process instrumentation (pH/ORP, conductivity and optical measurement), and owned the development of a new dual-channel pH/ORP product from specification and design through certification and production readiness.
- Developed embedded software for smart sensors and automation controllers, covering measurement, configuration, data handling and communication with mobile and web applications.
- Supported the sourcing and import of probes and components used across the sensor product line, and their integration into products.
- Led certification testing for the company's first officially BLE-certified device.

Automation and Industrial Communication

- Defined the technical parameters of and developed a new PROFINET-based controller product line on a shared embedded Linux platform, from a PROFINET I/O coupler to a full CODESYS PLC.
- Integrated the CODESYS runtime into the company's controller products and developed CODESYS (IEC 61131-3) projects and applications.
- Developed I/O module drivers and data exchange over PROFINET and Modbus, and integrated industrial interfaces (IO-Link, RS-485, CAN, 4-20 mA) across the product line.
- Redesigned the controllers' I/O architecture and rolled it out to 1000+ field devices, replacing legacy controllers.

HMI, Tools and Test Systems

- Developed HMI and user interfaces for devices and test systems, including web-based configuration, monitoring and update interfaces.
- Built CODESYS-based and PC-based test and calibration systems with automated test sequencing and data logging, shortening test and calibration time in series production.
- Developed APIs and simulation tools that let application and HMI developers build and test without physical hardware.

Production Handover, Quality and Field

- Defined production test procedures, manufacturing guidelines and quality criteria, and handed tests and calibration procedures over to the production team.
- Took part in first field trials, carried out post-delivery checks and resolved defects; led post-production quality assurance for the controller platform.
- Served as the technical escalation point for customer and field issues, running root cause analysis and training the field service team.
- Led a 6-month on-site feasibility study at customer dyehouses: interviewed process teams, collected samples, analyzed data with Python, SQL and Power BI, and presented the findings to management.

CORE COMPETENCIES

Project Coordination	Requirement and scope definition, product specification, schedule and task tracking, supplier and vendor coordination, customer interface, technical documentation
Industrial Communication	PROFINET IO, Modbus RTU/TCP, IO-Link, RS-485, CAN, 4-20 mA, MQTT
Automation & HMI	PLC programming (CODESYS, IEC 61131-3), CODESYS runtime integration, HMI and web UI development
Testing & Quality	Production test procedures, automated test and calibration systems, quality criteria, certification testing, root cause analysis
Instrumentation	pH/ORP, conductivity and optical measurement, analog front-end design, calibration
Software & Tools	C/C++, Python, embedded Linux, Docker, Git, Jira, Confluence, MS Office, Power BI, SQL

EDUCATION & LANGUAGES

Bachelor of Science, Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate

Driving licence Class B | Willing to travel internationally